Flora Shu-Qi Hamilton

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SUMMARY OF QUALIFICATIONS

Environmental data scientist with experience applying remote sensing, geospatial analysis, statistical modeling, and machine learning to ecological and natural-resource questions. Integrates and quality-controls field, lidar, satellite, and animal GPS data; develops analyses, models, maps, and tools using R, Python, Google Earth Engine, and GIS. Communicates research through technical reports, manuscripts, visualizations, and collaboration with scientists, agencies, land managers, and community partners. Experienced leading project teams and connecting analytical results to practical conservation and management needs.

PROFESSIONAL EXPERIENCE

Natural Resource Ecology Laboratory, Colorado State University, Fort Collins, CO (remote)
Research Associate II (9/24 – Present)

- Contributing to natural resource and ecological research, performing data analysis and visualization, statistical and machine learning modeling, and in-depth QA/QC of in-situ and remotely sensed data.
- Prepare research for publication and communicate findings to partners including CSU Extension, Colorado Parks and Wildlife, Colorado ranchers, and The Nature Conservancy.

NASA DEVELOP, Analytical Mechanics Associates, Inc.
Project Lead (Goddard Space Flight Center, 9/24-11/24)

- Led a 4-person team utilizing NASA remote sensing data to map oak tree mortality risk in Illinois for the Morton Arboretum. Produced maps, models, and tools using ArcGIS, R, Python, and the Climate Engine API. Coordinated with stakeholders to align project outcomes with current oak management strategies.

Geospatial Analyst (Colorado State University, 6/24-8/24) | Technical Paper

- Assessed snowpack-wildfire relationships for water utility management in Colorado.
- Evaluated Lidar and field data, developed Random Forest models, and contributed analysis to tech report.

Cary Institute of Ecosystem Studies, Millbrook, NY
Data Jam Data Intern (12/22–7/23)

- Worked with scientists to create datasets, worksheets & metadata for Hudson Data Jam Competition.
- Assisted in classroom instruction, communicating data science and environmental science principles.

SOSV and Orbit Startups, Taipei, Taiwan
Investment Specialist (6/21–2/22)

- Sourced, researched, and met with potential start-ups in SEA region, conducting interviews with general partner for investment. Researched and wrote memos for add-on investments of global VC with 1.3+ AUM.
- Advised, supported, and reviewed investment, financial, and sales materials for portfolio companies.

Integrative Economics, LLC, Portland, OR
Independent Contractor (7/22–10/22) & **Product Development Intern** (3/20–8/20)

- Identified high-impact areas for pollinator habitat restoration in Oregon using cluster analysis in QGIS.
- Analyzed COVID's economic impact on the domestic blueberry industry for the High-Bush Blueberry Council.

EDUCATION

Master of Environmental Data Science '24, Bren School of Environmental Science & Management, UCSB
Bachelor of Arts in Mathematics and Environmental Studies '21, Bowdoin College, Brunswick, ME

RELEVANT SKILLS

Technical: R, Python, Git & GitHub, Google Earth Engine, ArcGIS, Markdown, SQL, HTML, CSS, database management, data analysis, geospatial analysis, statistical analysis, machine learning, dashboard development, Google & Microsoft Suite | **Languages:** English, Mandarin Chinese (Fluent), French (Working proficiency)